

Private Cloud Computing

with UEC

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A cloud is a cluster of servers, where servers can be added and removed, on demand. The ability of the cloud to match computing capacity with demand (referred to as elasticity), allows a cloud to offer high performance and scalability, while maintaining high server utilisation, since hardware resources are more closely related with demand and workloads. A cloud thus eliminates the need to maintain excess server capacity.

Clouds are typically of two kinds—public clouds and private clouds. In a public cloud, the servers are owned and managed by a hosting provider (for example, Amazon, Slicehost, etc), and are located at the hosting provider's site. In a private cloud, the servers

are owned and managed by the company/organisation using it, although they can be located either at the company's site or co-located at a hosting provider's site.

Ubuntu Enterprise Cloud (UEC), available in Ubuntu Server 9.10 (Karmic Koala), offers the functionality for both private and public clouds. UEC allows for the creation of a private cloud consisting of servers running Ubuntu Server 9.10. UEC also allows the easy creation of Ubuntu server images that can be deployed on Amazon EC2 public clouds. UEC is built using a number of open source technologies, which include KVM, QEMU, libvirt and Eucalyptus. This article describes the key concepts of these technologies, and shows how UEC can be used to build a private cloud.